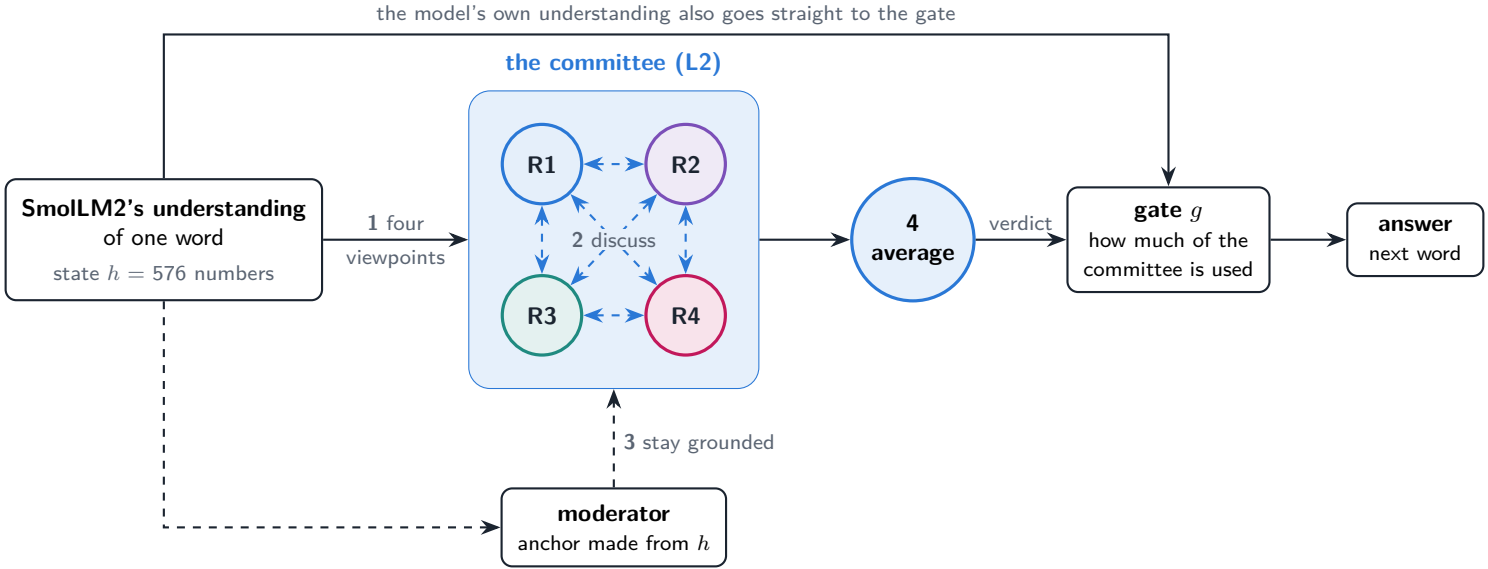
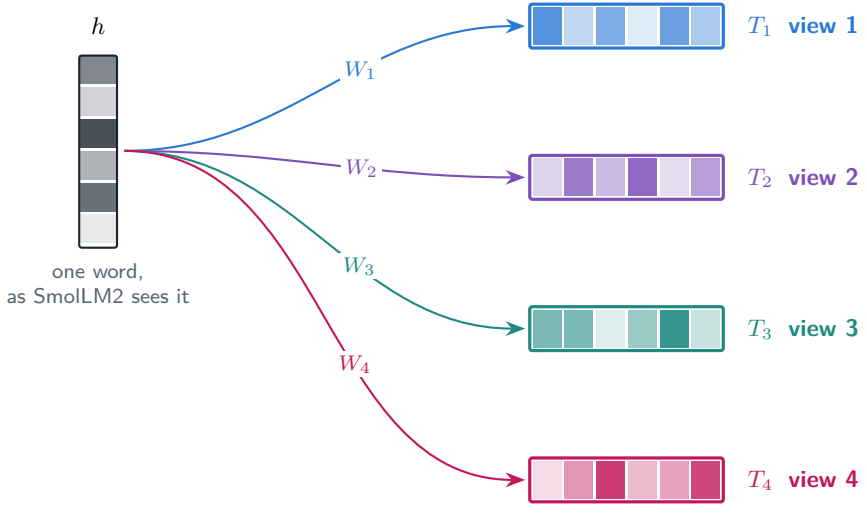
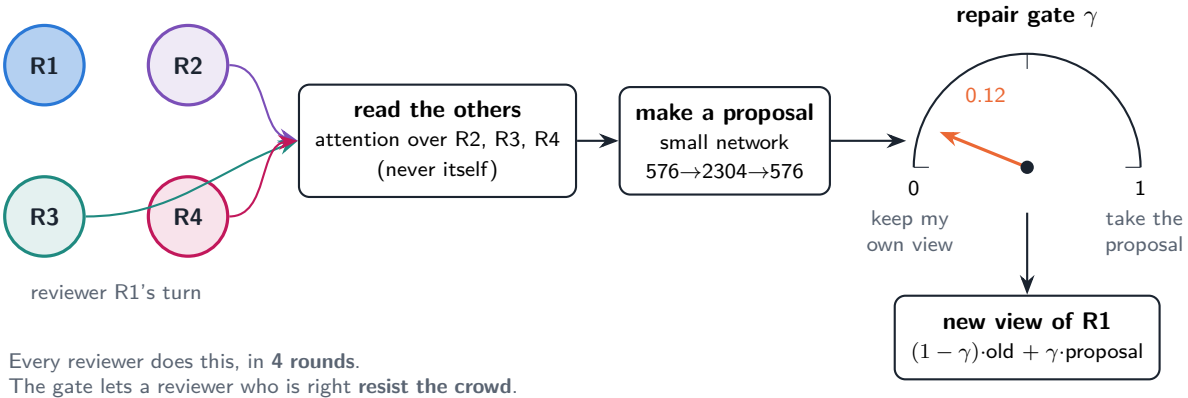

the model's own understanding also goes straight to the gate
the committee (L2)
SmolLM2's understanding
of one word
state $h = 576$ numbers
1 four
viewpoints
R1
R2
R3
R4
2 discuss
4
average
verdict
gate g
how much of the
committee is used
answer
next word
3 stay grounded
moderator
anchor made from h

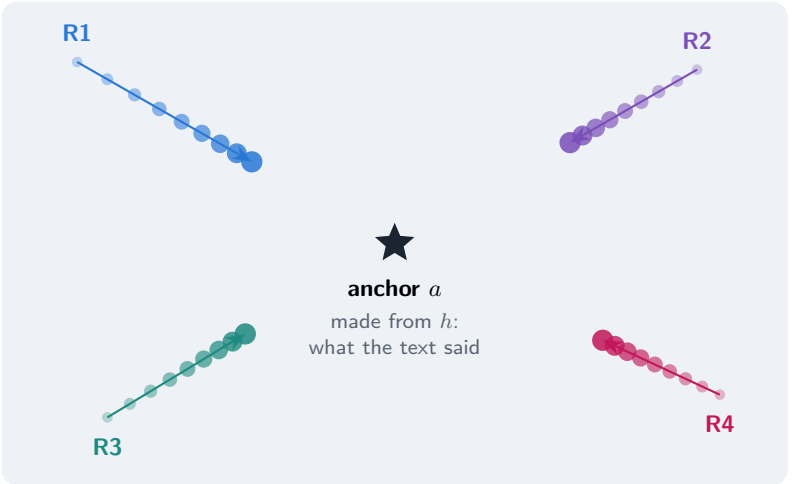


Four learned transformations $T_k = W_k\, h + b_k$ turn the same word into four different views.

They are built to start **clearly different** (orthogonal), so there is something to discuss.

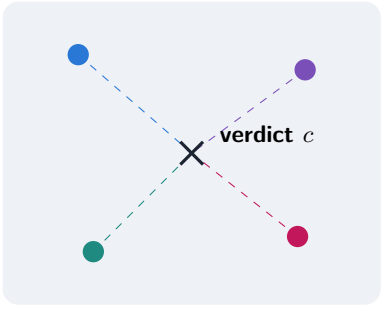


Every reviewer does this, in **4 rounds**.
The gate lets a reviewer who is right **resist the crowd**.



8 small steps towards the anchor:
$T_k \leftarrow T_k - \eta\, s_k \odot (T_k - a),\ \ \eta = 0.0625.$

Each reviewer has its **own pull** strength s_k, so they move by different amounts and **do not** collapse into one point.



gate g: how much of the verdict is used

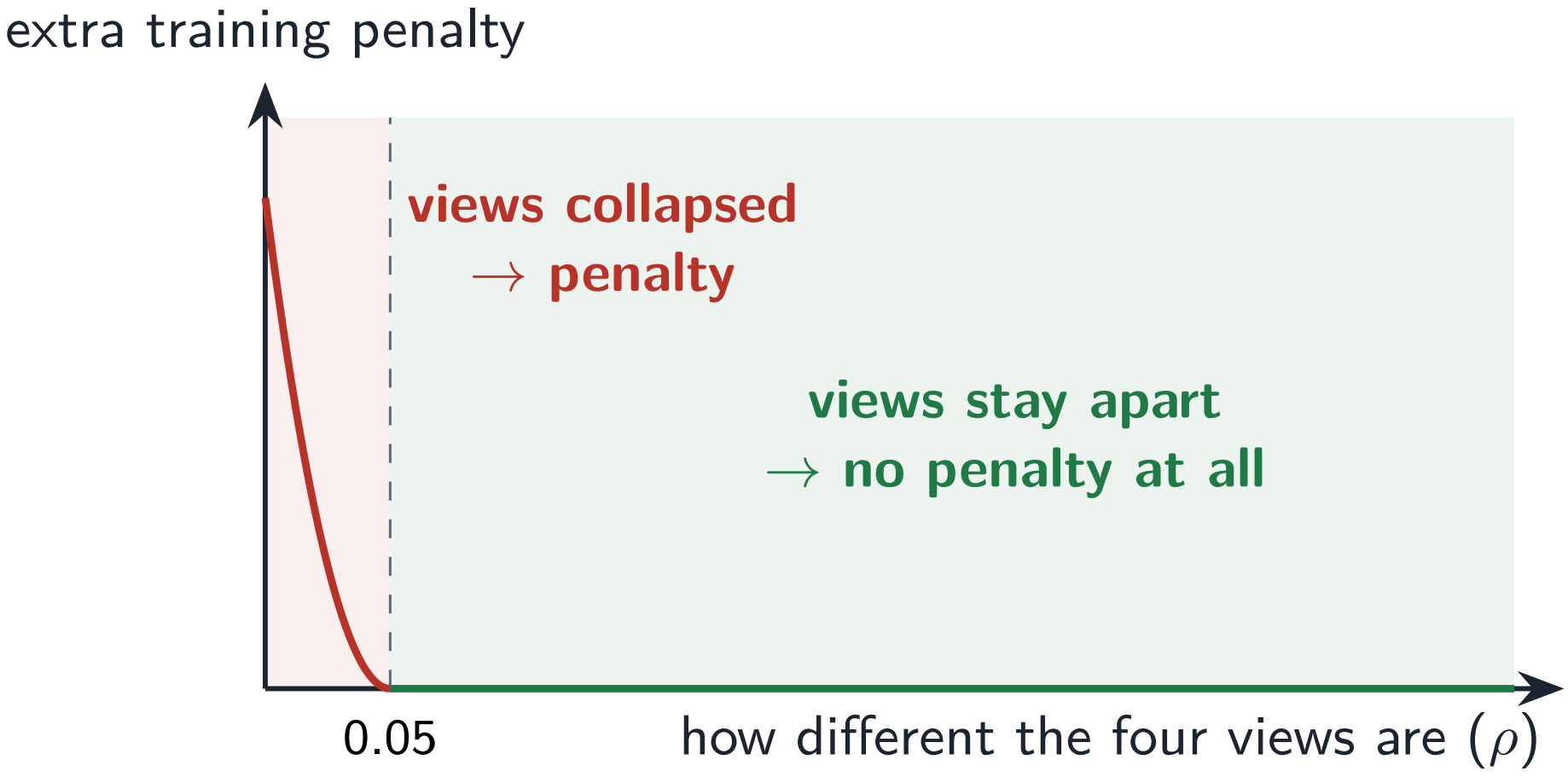
after training: $g \approx 0.027$
$\approx$ 3 % committee

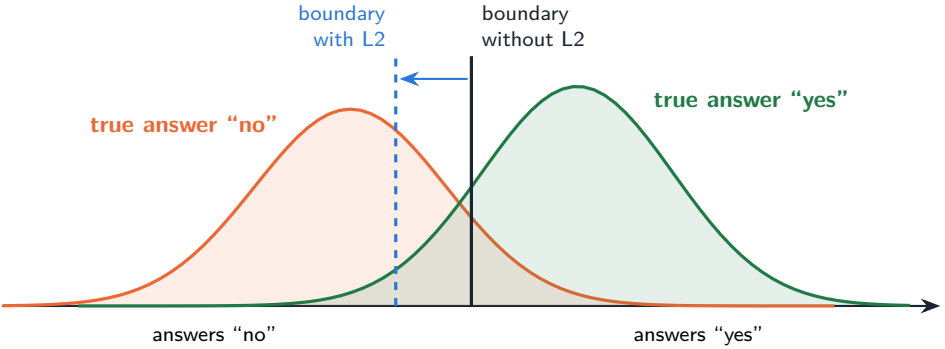
$g = 0$
only SmolLM2
(start of training)

$g = 1$
only the
committee

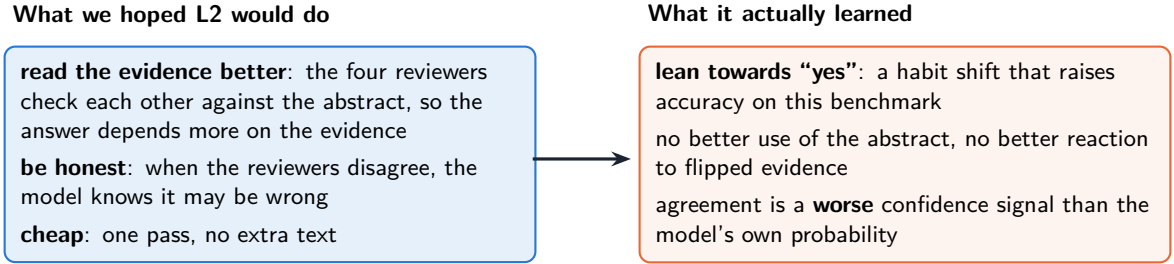
final state $= h + g \cdot (\text{verdict} - h)$

4 the four views are averaged


extra training penalty
views collapsed
→ penalty
views stay apart
→ no penalty at all
0.05
how different the four views are (ρ)



Moving the boundary left turns some answers into “yes”: more yes-questions right ($+11.0$), more no-questions wrong (-11.8). With 552 yes- and 338 no-questions, accuracy rises by 2.07 points although nothing about reading the evidence changed. (Curves are an illustration; the numbers are measured.)


What we hoped L2 would do
read the evidence better: the four reviewers check each other against the abstract, so the answer depends more on the evidence
be honest: when the reviewers disagree, the model knows it may be wrong
cheap: one pass, no extra text
What it actually learned
lean towards "yes": a habit shift that raises accuracy on this benchmark
no better use of the abstract, no better reaction to flipped evidence
agreement is a worse confidence signal than the model's own probability